Part I: pre-midterm-2 content.

Problem 1. *Provide a parameterization $\vec{r}(t)$ of the curve C formed by the intersection of the circular cylinder of radius 1 about the z-axis, and the plane $z = y$. The parametrization must start at $\vec{r}(0) = (1, 0, 0)$ and end at the point $(0, 1, 1)$.*

Problem 2. *Sketch vector arrows to depict the vector field*

$$\vec{v} = -y\hat{\boldsymbol{i}} + x\hat{\boldsymbol{j}}.$$

Problem 3. *Evaluate $\iiint_T (x+y+z)\, dxdydz$ where T is the volume of the first octant inside the cone $x^2 + y^2 \leq 4z^2$, $z \in [0, 2]$, as shown below.*

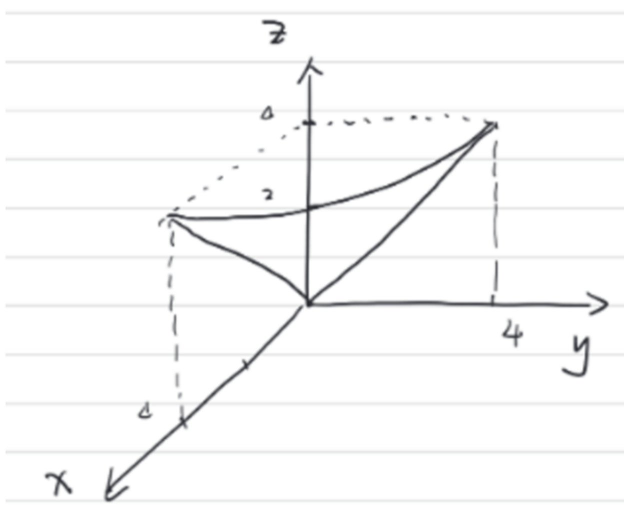

Figure 1: Region T.

Problem 4. *Verify Stokes' Theorem for*

$$\vec{F} = [y^3, -x^3, 0],$$

and S is the surface

$$x^2 + y^2 \leq 1, \quad z = 0.$$

Hint: To save time, you may directly use, if needed, the following facts:

1. *S can be parameterized as $x = u\cos v$, $y = u \sin v$, for $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$, and in this set up, $\vec{N} = \vec{r}_u \times \vec{r}_v = u\hat{k}$.*
2. *The boundary ∂S, can be parameterized as $x(t) = \cos(t)$, $y(t) = \sin(t)$, $0 \leq t \leq 2\pi$.*
3. *The expression $\sin^4(t) + \cos^4(t)$ can be simplified to $\frac{3}{4} + \frac{1}{4}\cos(4t)$.*

Part II: post-midterm-2 content.

Problem 5. ***Conceptual Questions.***

1. *While modeling 1D standing waves on a perfectly elastic string initialy stretched to some length L, what is the expression for c in terms of the tension T and μ — the uniform mass per unit length?*

2. *For the Bessel function of the first kind of zeroth order $J_0(s)$, what is the value of $J_0(\alpha_5)$?*

3. *The solution to the heat equation for a very long bar is written as*

$$u(x,t) = \int_0^\infty \left[A(p)\cos(px) + B(p)\sin(px)\right] \mathrm{e}^{-c^2p^2t}\, dp.$$

   For the initial condition $u(x,0) = f(x)$, write down an expression for $A(p)$.

Problem 6. *Solve the following PDE using Laplace Transformation:*

$$\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial t} = x,$$

with $w(x,0) = 1$ and $w(0,t) = 1$.

You may use the fact, without proof, that the solution to $\dfrac{\partial W}{\partial x} + xsW = x + \frac{x}{s}$ is

$$W = C(s)\mathrm{e}^{-sx^2/2} + \frac{1}{s^2} + \frac{1}{s}.$$

Problem 7. *Use separation of variables to find a solution to the PDE*

$$\frac{\partial w}{\partial x} + 2x\frac{\partial w}{\partial t} = 0.$$

No initial or boundary conditions are given.

Problem 8. *Solve the 2D wave equation $u_{tt} = c^2\nabla^2 u$ on a circular membrane of radius R, where the deformation u is radially symmetric, i.e., $u = u(r,t)$, and satisfies the initial condition $u(r,0) = 2.11J_0\left(\frac{\alpha_5}{R}r\right)$. What are the radii of the (inner) nodal circles for the harmonics? What are the corresponding harmonic frequencies?*